

duolingo



Duolingo Redesign Group 15: Language Inequalities

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Problems with Duolingo



Limited real-world interaction: No native speaker integration for practice.



Lacks immersion: Doesn't use AR to enhance everyday learning.



Isolated experience: No peer learning/community engagement within the app.

Problems

Lack of in-depth
grammar explanations

W70

Repetitive and
demotivating lessons

W70

Lack of immersive
real-world practice

W70

Ineffective tracking
of progress

W70

Suggestions

Gamified learning with
personalized storylines

W70

Personalized AI assistant
for spaced repetition

W70

Live conversation with
a native speaker

W70



AI-generated real-
world scenarios

W70

Personalized AI assistant
for spaced repetition

W70

Tailored Grammar
Lessons Prior to Starting
Placement Tests

W70



Culturally Authentic
Conversation Scenarios

W70

W70

More Social and
Competitive Features

W70

Augmented reality-based
learning

W70



Daily News & Pop
Culture Lessons

W70

Social Media
Integration

W70



Better Motivation with
Encouraging Duolingo
Learning

W70



Features Added

► Feature 1: Social Learning Community Hub

- Reasoning
 - Stay motivated by joining in on challenges with friends or peers.
 - Help fill users' desire for peer interaction and collaborative learning, which Duolingo currently lacks.
 - Makes learning a language more immersive and enjoyable.

► Feature 2: Augmented Reality Learning Based System

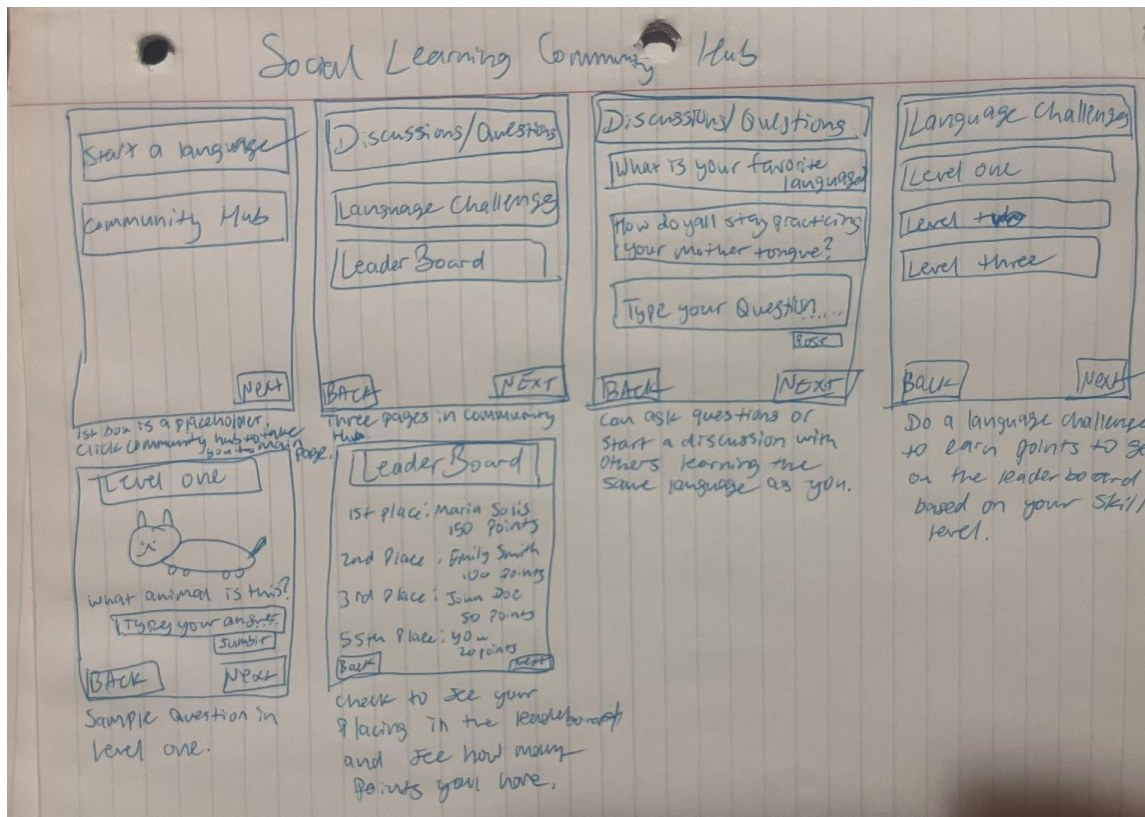
- Reasoning:
 - Fills the gap of real-world immersion that Duolingo currently lacks.
 - Allows learners to interact with their physical surroundings and connect vocabulary to real-life objects.
 - Helps users in foreign environments learn contextually by translating objects they encounter every day.

► Feature 3: Live Conversation with Native Speaker

- Reasoning:
 - Stay motivated by applying the vocabulary you learn in real world contexts
 - Get immediate feedback on mistakes in pronunciation, word usage, etc.
 - Allow users to practice with a human being and have a more natural seeming conversation

Feature 1: Social Learning Community Hub

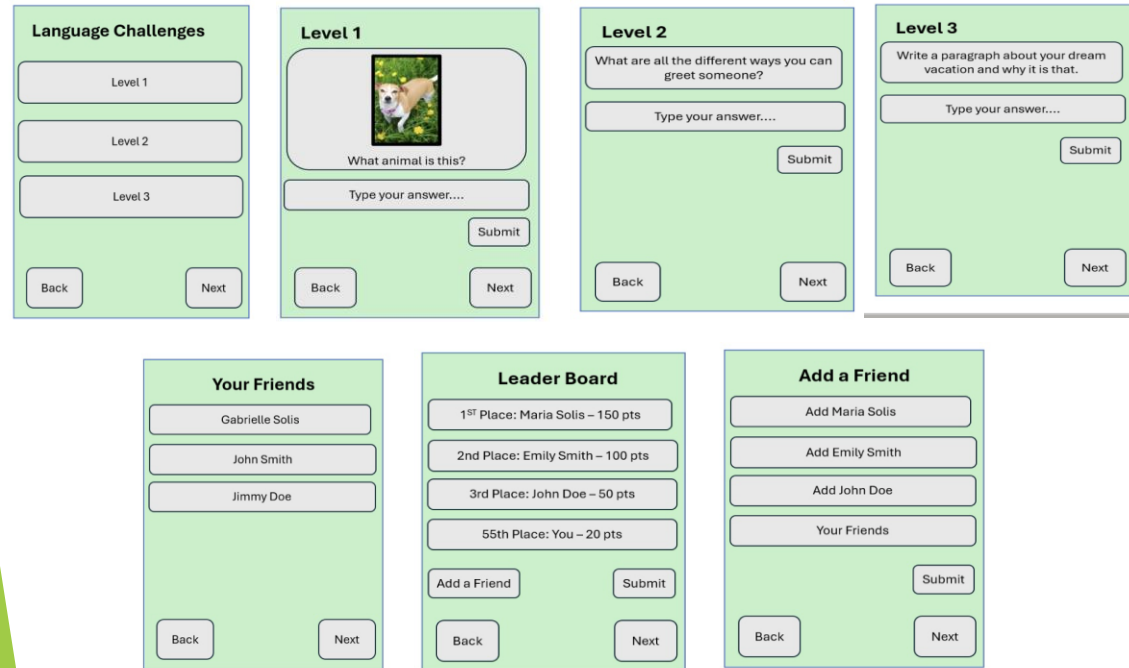
Feature 1



- Idea: Forum-style hub with language challenges, discussion, and competitive minigames.
 - Inspiration: Social media and online game communities like Reddit and Discord.
 - User quote: "I'd love to play games with friends in Spanish—it'd be way more fun."
 - User Feedback: Users liked the idea of interacting with other learners and seeing a leaderboard.
- Struggled to distinguish UI elements (e.g., no clear labels for different sections like "Discussion" or "Leaderboard").

Testing Phases Feature 1

Low Fidelity Prototype



Mid Fidelity Prototype

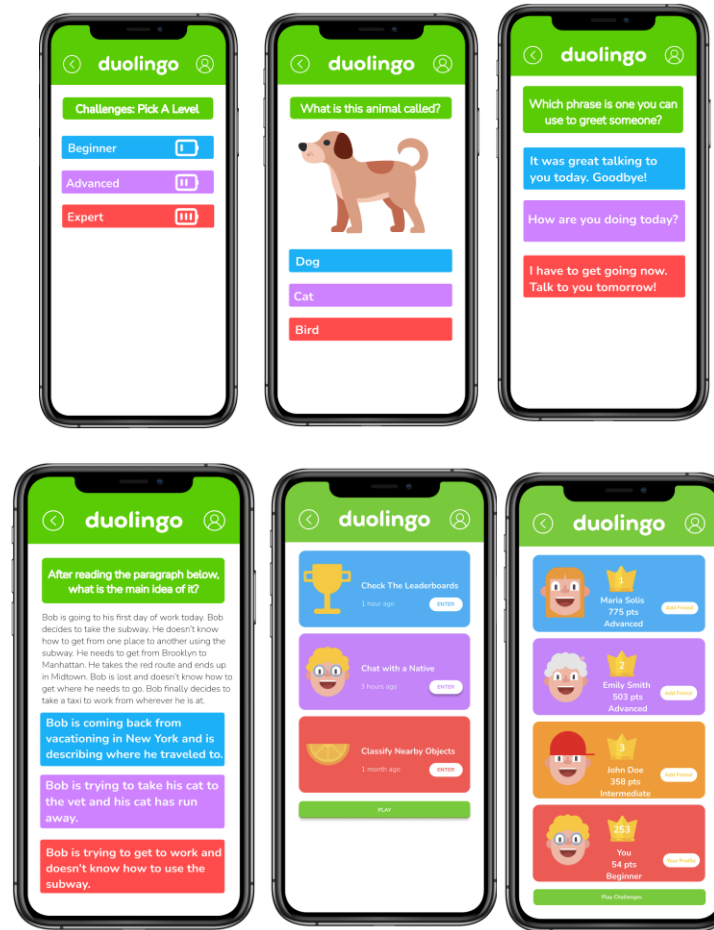


Focused on layout and feature structure using simple wireframes. Prioritized defining user flow across the community hub feature. Helped us identify early usability concerns and get quick feedback on functionality.

Added visual elements and mobile frame to simulate actual usage. Improved navigation and feature clarity.

Testing Phases Feature 1

High Fidelity Prototype



Polished visual design with finalized layout and branding consistent with Duolingo. Enabled full interaction with all three features—object translation, live chat with native speakers, and challenge-based social hub.

Learning Science Principle Feature 1



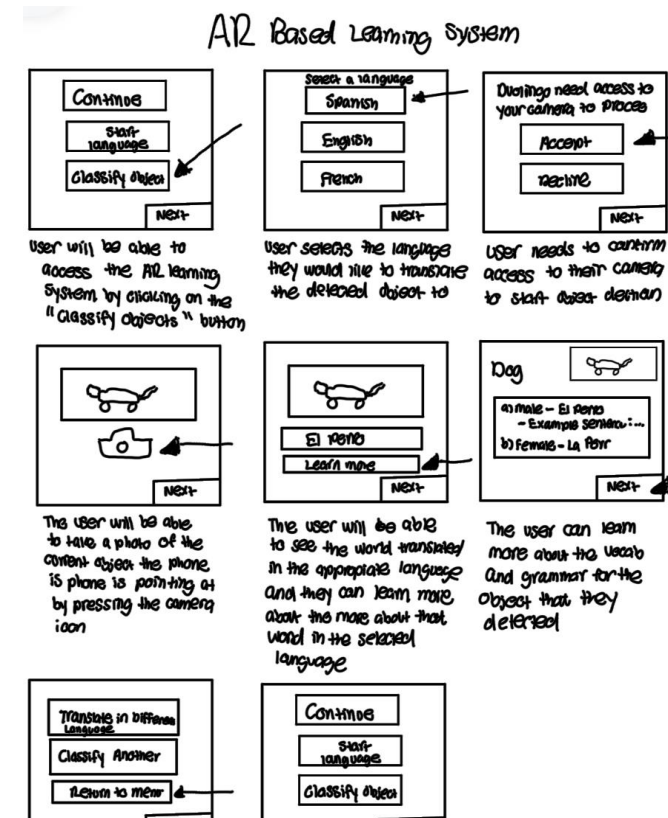
- ▶ Social Constructivism: Learning through social interactions.
- ▶ Peer Scaffolding: Learners support each other's growth (Vygotsky).
- ▶ Primary Reading Tie-In: "Closing the Opportunity Gap" - Importance of community for motivation and belonging.

Feature 2: Augmented Reality Learning Based System

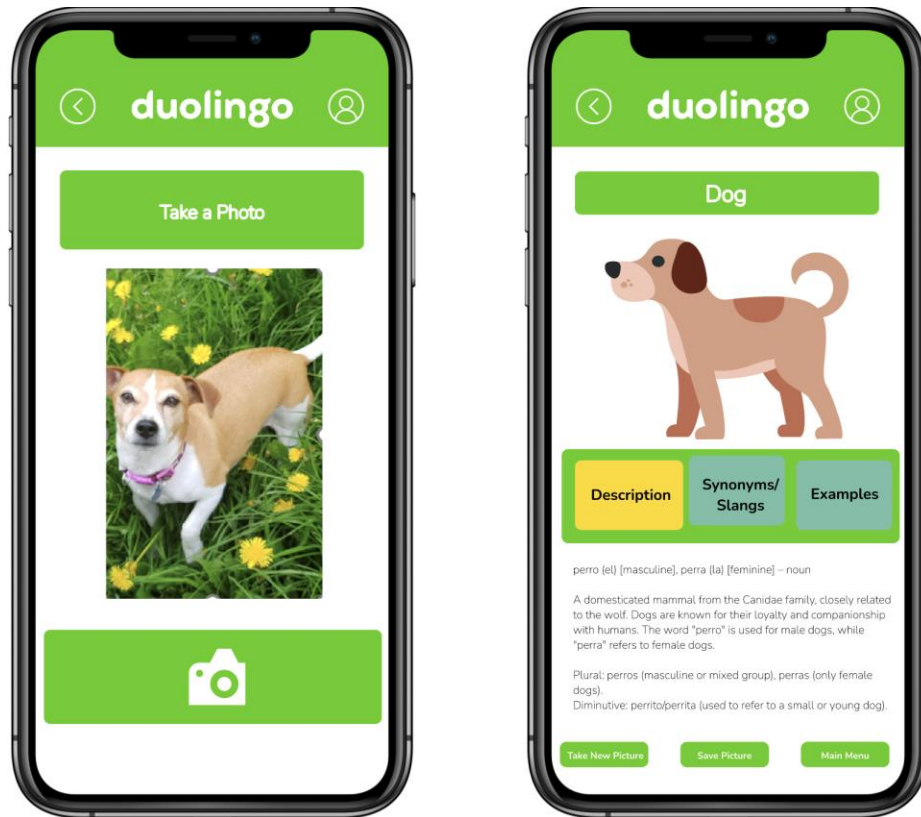
Feature 2

- Idea: An AR feature where users can point their phone's camera at real-world objects to see the name translated in their target language. This turns everyday surroundings into immersive vocabulary lessons.
- Inspiration: Inspired by Google Lens, this tool integrates object classification with translation to enable dynamic, context-aware language learning.
- User Quote: "Sometimes I wish I could just look at something and instantly know what it's called in French."
- User Feedback: Users appreciated the interactive concept of scanning objects.

Had trouble telling which UI components were clickable (e.g., buttons vs. text).



Learning Science Principle Feature 2



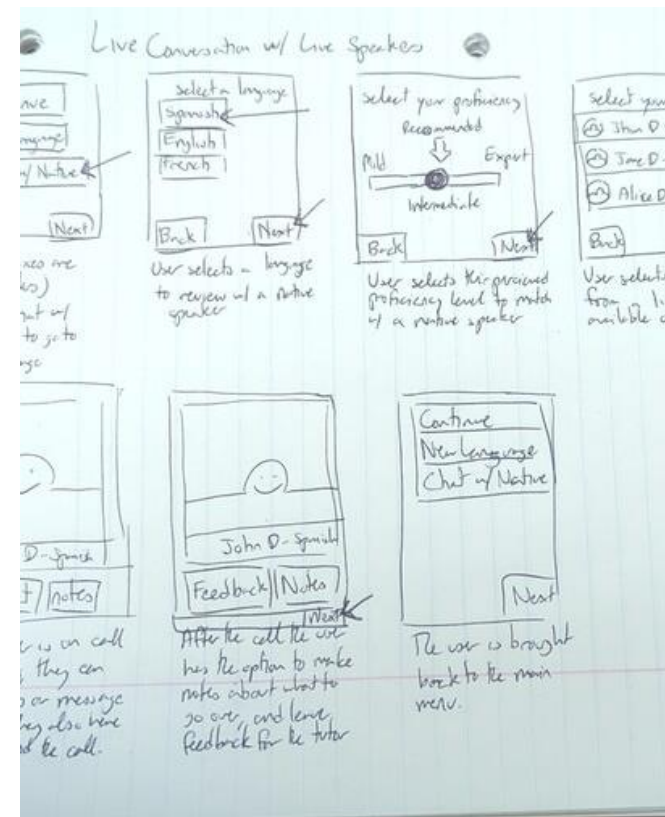
- ▶ **Situated Cognition:** Learning is most effective when it is contextually grounded. By using AR to classify real-world objects in a target language, users connect new vocabulary to physical, lived experiences
- ▶ **Distributed Cognition:** The AR feature integrates tools like the camera, image recognition, and on-screen feedback as part of the learner's cognitive system. This aligns with enhancing memory and reducing cognitive load.
- ▶ **Primary Reading Tie-In:** "Closing the Opportunity Gap" discusses about designing for cultural relevance and embracing the learners' native environments. This AR feature allows for diverse linguistic backgrounds by treating users' surroundings and languages as assets rather than deficits

Feature 3: Live Conversation with Native Speaker

Feature 3

- ▶ Idea: A video camera call integration feature where a language learner can connect with a native speaker for that language
- ▶ Inspiration: Random video call sites such as Omegle
- ▶ User quote: “It'd be a lot more meaningful if I had a place to apply what I was learning with another person”
- ▶ User Feedback: Users liked seeing tutor profiles but were confused about selection steps.

Unsure what to do after selecting a tutor — no clear guidance or next steps.



Learning Science Principle Feature 3



- ▶ Social Constructivism: Learning through social interactions.
- ▶ Primary Reading Tie-In: "How People Learn - Chapter 6" - Utilizes extrinsic motivation.
- ▶ Procedural (applied) Knowledge: Learners often enter situations with prior knowledge (Hailikari, 2006)

Testing Phases Summary

Low Fidelity Prototype



Mid Fidelity Prototype

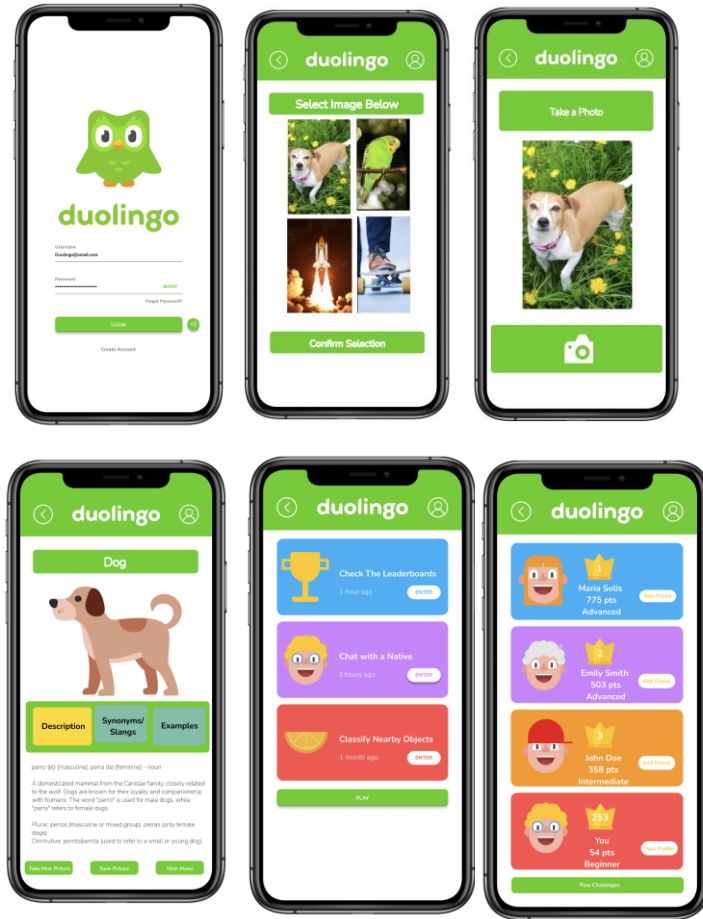


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Testing Phases Summary

High Fidelity Prototype



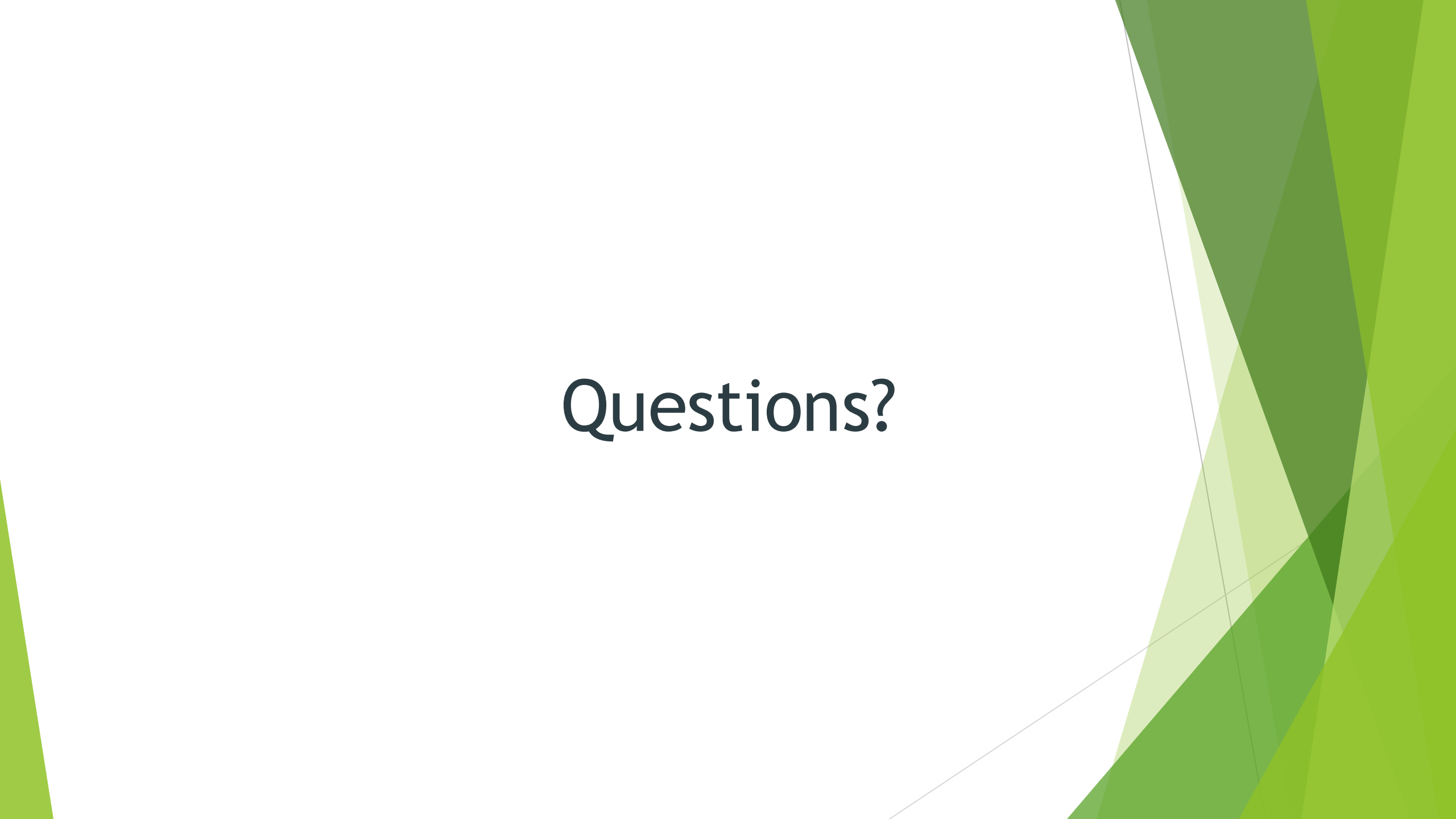
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Video Demo of Full Prototype



Summary of Changes

- ▶ Added a forum-style community hub where users can post questions, interact with other learners, and engage in language challenges
- ▶ Added an AR-based tool that lets users scan real-world objects and get translations in their target language, enhancing contextual learning
- ▶ Added a video calling feature that pairs users with native speakers for live language practice, helping improve fluency

The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern, layered effect on the right side of the slide.

Questions?